

Shuhan Zheng

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Summary

M.Eng graduate from the University of Toronto. Demonstrated strong experience in streamlining complex analytical problem solutions via approaches grounded in first principles. I bring expertise in physics-aware software development, especially those pertaining to nonlinear behaviors in dynamical systems, or those that require intensive computation. Multidisciplinary engineering, large data-driven functions, and complex cross-system work are also in my forte. Supplementary skills in electronics, mechanical prototyping, and systems-level analysis.

Education

University of Toronto Institute for Aerospace Studies (UTIAS) 2024–2026
M.Eng. in Aerospace Engineering
Coursework: Spacecraft Dynamics and Control, Control Theory, Computational Fluid Dynamics, Project Management, Operations Management, etc.

University of Toronto 2020–2024
Honours B.Sc. with double majors in Physics and Astronomy & Astrophysics

Experience

Research Assistant / Supervised Research Project Spring 2024–Present
University of Toronto Institute for Aerospace Studies, Aerospace Mechatronics Lab

- Abstract accepted by AMOS 2026 as **first author**: *Dichotomizing Temporarily Bound Objects Using Capture States*.
- Large data-enabled Space Domain Awareness study powered by techniques such as statistical modeling and probabilistic neural networks, enabling rapid and early detection of mission-viable asteroids.

Research Assistant 2023–2024
University of Toronto, Near Field Cosmology Group

- Built a **turnkey solution** for automatic retrieval, sanitizing, and cataloging of stellar data, with a portable interface for machine-learning model ingestion.
- Project adopted by the Near Field Cosmology Group; the resulting databases supports subsequent publications.

Research Assistant Intern Summer 2023
China Spallation Neutron Source, Chinese Academy of Sciences

- **Independently** developed a Python-based deterministic simulation tool for generating synthetic polarized-neutron images of magnetized samples.
- Computational time reduced from days to hours compared to conventional Monte Carlo methods.

Space Systems Member 2021–2023
University of Toronto Aerospace Team

- Implemented and tested image-correction algorithms for hyperspectral remote-sensing data in collaboration with the optics team. The algorithms will support the upcoming FINCH **microsatellite mission**.
- Co-authored team design papers for the 2022 and 2023 Small Satellite Conferences as a member of the FINCH mission team.

Vice-President, Internal and External Affairs 2023–2024
University of Toronto Physics Student Union

- Managed the organization, logistics, and coordination of outreach, fundraising, social, and academic events.
- Acted as a liaison between faculty and students.

OPS & Safety Operator 2026
Launch Canada

- Responsible for assisting the launch OPS and Safety team, enforcing launch pad clearance and compliance to launch safety regulations.

Projects

AMAT - Agentic Mission Analysis Toolbox
Independent Software Project (Link: [GitHub](#))

- AI harness and toolbox for reliable application in spaceflight mission design.

- End-to-end solution to convert user intent into real, mission-ready spacecraft flight plans, with interactive visualization capabilities for human-in-the-loop verification and validation.
- Compatible with a range of missions, from LEO to cis-lunar and beyond.

GMAT Automation for Asteroid Trajectory Prediction

Astrodynamics Course Project (Link: [GitHub](#))

- Monte Carlo simulation of real asteroid trajectories for prediction of future Earth with uncertainties.
- Earth encounter predictions verified consistent with published journal literature.

Catalog Acquisition and Compilation tool

Astronomy Capstone Project (Link: [GitHub](#))

- Autonomous data retrieval, sanitizing, and management pipeline with user interface API.
- Scalable scientific database solution designed for hierarchical and relational data.

Polarization-modulated Optical Communication Demonstrator

Electronics Course Project

- Designed and built a prototype for serial transmission of binary information via laser link.

Publications

- Miles, A., Maranto, D., Xu, S., et al., **Zheng, S.**, et al. (2022). *FINCH: A Blueprint for Accessible and Scientifically Valuable Remote Sensing Satellite Missions*. Small Satellite Conference, Weekend Session 7: Science/Mission Payloads – Research & Academia II, Utah State University, Logan, UT. <https://digitalcommons.usu.edu/smallsat/2022/all2022/88/>
- Vyse, I., Dagli, R., Chadha, D. V., Ma, J. P., Chen, H., et al., **Zheng, S.**, & Nguyen, K. (2024). *Beyond the Visible: Jointly Attending to Spectral and Spatial Dimensions with HSI-Diffusion for the FINCH Spacecraft*. Small Satellite Conference, Session X: Ground Systems, Utah State University, Logan, UT. <https://digitalcommons.usu.edu/smallsat/2024/all2024/121/>

Skills

Programming and Analysis: MATLAB, LaTeX

Aerospace and Simulation Tools: GMAT, STK, COMSOL Multiphysics, SolidWorks, CFD